COURSEWORK ASSIGNMENT SUBMISSION

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Subject: Software Engineering & Project Management **Day:** Day 26 — Project Documentation, Presentation & Jira Workflows

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Section 8: Classroom Practical — Jira-Style Project Workflow Plan

Structured Jira backlog hierarchy, issue linking, and board workflow planning for the College Event Management System (CEMS).

1. Jira Issue Hierarchy Table

Issue Key	Issue Type	Summary	Priority	Linked Dependencies	Completion Evidence / DoD
CEMS-EP-01	Epic	Student Event Registration Management	High	None	All child stories deployed and verified.
CEMS-ST-101	Story	View event details and real-time seat availability	High	Child of CEMS-EP-01	Catalog UI displays dynamic seat counter.
CEMS-ST-102	Story	Register for available event atomically	Highest	Blocks CEMS-ST-103	Seat reserved; confirmation token stored.
CEMS-TK-201	Task	Implement database row-level locking query	Highest	Sub-task of CEMS-ST-102	`FOR UPDATE` query passes unit tests.
CEMS-TK-202	Task	Build responsive mobile registration form UI	High	Sub-task of CEMS-ST-102	Form renders cleanly at 360px mobile width.
CEMS-BG-301	Bug	Full event accepts registration under concurrency race	Highest	Relates to CEMS-ST-102	Defect reproduced, fixed, regression passed.

2. Sprint Workflow Board States

Standard Workflow Columns: [BACKLOG] -> [SELECTED FOR SPRINT] -> [IN PROGRESS] -> [CODE REVIEW / PR] -> [TESTING / QA] -> [DONE].

Transition Rule: An issue can only move from [CODE REVIEW] to [TESTING] once GitHub pull request has 1 approved review and CI automated build passes.

Section 12: Homework — Project Documentation Packet & Presentation Outline

Complete project handover documentation and executive presentation plan for campus stakeholders.

1. Five-Slide Project Presentation Outline

- **Slide 1 (Executive Summary & Problem Statement):** Title: Modernizing Campus Event Management. Outlines legacy manual registration chaos, seat allocation disputes, and operational goals of CEMS.
- **Slide 2 (Requirements & System Scope):** Summary of 12 functional requirements (catalogs, atomic booking, QR check-in) and non-functional targets (sub-1.5s latency, 99.5% uptime).
- **Slide 3 (Architecture & Database Schema):** Diagram of 3-Tier Layered Architecture (React SPA -> Express REST API -> PostgreSQL 16) with Redis asynchronous message queues.
- **Slide 4 (Verification Evidence & Defect Triage):** Test execution dashboard showing 98% pass rate, zero critical defects, and automated JMeter concurrency benchmark results.
- **Slide 5 (Project Milestones & Future Roadmap):** Demonstrates Phase 1 MVP completion on schedule and presents roadmap for Phase 2 (payment gateway, waitlist automation).

2. Reflection on Project Traceability

Maintaining explicit linkages between Jira user stories, Git feature branches, commit hashes, and automated test cases prevents orphaned code, ensures that every single line of written software serves an approved business requirement, and allows auditors to instantly verify test evidence for any production release.

DARSH